

Erdős Problem 915: Cheat Sheet (front)

one line per bound, page numbers into main.pdf

conjectural

PROVED

exact value established

PARTIAL

partial or

OPEN

general value open

Simple graphs

$\ell_m(n)$ edge

PROVED

UP $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$, every n : Gomory–Hu tree, at most $n-1$ edges cross one tree cut and the rest cross two, against total cut capacity $(m-1)(n-1)$ [p. 37–39]

LO $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$ $n \geq m \geq 2$: hub-and-spokes graph, spoke degree $\leq m-2$ [p. 40]

$k_m(n)$ vertex

PARTIAL

UP $= \left\lfloor \frac{m(n-1)}{2} \right\rfloor$ $m \leq 4, n \geq m$: the edge bound does not transfer ($\kappa \leq \lambda$ is the wrong direction), equal anyway up to $m=4$ (Bárfai, Bollobás, Leonard) [p. 6]

LO $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$ $n \geq m \geq 2$: same hub-and-spokes [p. 40], and Sørensen–Thomassen’s value $\lfloor 8n/3 \rfloor - 4$ overtakes it at $m=5, n \geq 14$ [p. 6]

$\ell_m^{\text{dir}}(n)$ arc

OPEN

UP $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$ $m=2$: a feasible digraph is its own reachability skeleton, and a skeleton has at most $M(n)$ arcs [p. 47,51]

LO $M(n)$: bidirected hub [p. 46] or one-way bipartite digraph [p. 46]

$k_m^{\text{dir}}(n)$ vertex

OPEN

UP $M(n)$ $m=2$: an arc outside the skeleton has a second route, internally disjoint from the arc itself [p. 47,50]

LO $\lfloor n^2/4 \rfloor$: one-way bipartite digraph [p. 46]

Multigraphs

$L_m(n)$ edge

PROVED

UP $(m-1)(n-1)$, every n, m : Gomory–Hu, $n-1$ cuts each of capacity $\leq m-1$, every edge crosses one [p. 40]

LO $(m-1)(n-1)$: any spanning tree at multiplicity $m-1$ [p. 40]

$K_m(n)$ vertex

PARTIAL

UP $2(n-1)$ $m=3$: incidence-rank lemma at $r=2$ [p. 70,75]

LO $(m-1)(n-1)$: tree thickened to multiplicity $m-1$, no detours, so the whole cap goes to multiplicity [p. 75]

$L_m^{\text{dir}}(n)$ arc

PROVED

UP $(m-1)M(n)$, every n, m : peel a reachability skeleton off $m-1$ times [p. 58–59]

LO $(m-1)M(n)$: thickened bidirected tree or one-way bipartite digraph [p. 58]

$K_m^{\text{dir}}(n)$ vertex

OPEN

UP $(m-1) \lfloor n^2/4 \rfloor + 4(m-1)^2(n-1)$ far from tight at small n [p. 83]

LO $n(n-1)(m+1-n)$ proposed, $2 \leq n \leq m$: thickened complete digraph, 24 at $n=3, m=6$ [p. 84]

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Hypergraphs

$\ell_m^{(r)}(n)$ **edge**

PARTIAL

UP $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$: cut induction on trace hypergraphs (Berge forbids the graph reduction) [p. 64]
LO $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ $m-1 \leq \binom{n-2}{r-2}$: star hypertree or degree-capped hub [p. 65–66]

$k_m^{(r)}(n)$ **vertex**

PARTIAL

UP $\left\lfloor \frac{n-1}{r-1} \right\rfloor$ $m=2$: incidence graph is a forest ($r|E| \leq n + |E| - 1$) [p. 67]
LO $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ $r \geq 3, m-1 \leq \binom{n-2}{r-2}$: degree-capped hub [p. 69]

$\ell_m^{\text{dir}(r)}(n)$ **arc**

OPEN

UP $\frac{(m-1)n(n-1)}{r-1}$ not tight for large n : tail–head pair counting [p. 85]
LO $n \cdot \min(m-1, \binom{n-1}{r-1})$ $n \geq r$: bounded outgoing hyperedge count [p. 86]

$k_m^{\text{dir}(r)}(n)$ **vertex**

OPEN

UP $\frac{(m-1)n(n-1)}{r-1}$: same pair counting, one-step routes are also internally disjoint [p. 87]
LO $n \cdot \min(m-1, \binom{n-1}{r-1})$ $n \geq r$: same construction, the outgoing cut ignores interior vertices [p. 86]

Multihypergraphs

$L_m^{(r)}(n)$ **edge**

PARTIAL

UP $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$, unchanged by repetition: same cut induction, repetition-blind [p. 64]
LO $\frac{(m-1)(n-1)}{r-1}$ $(r-1) \mid (n-1)$: star hypertree, every hyperedge repeated $m-1$ times [p. 65]

$K_m^{(r)}(n)$ **vertex**

PARTIAL

UP $\left\lfloor \frac{2(n-1)}{r-1} \right\rfloor$ $m=3$: incidence-graph cycle rank via cut vertices and separating pairs [p. 70]
LO $\frac{(m-1)(n-1)}{r-1}$ $(r-1) \mid (n-1)$: repeated star hypertree [p. 69]

$L_m^{\text{dir}(r)}(n)$ **arc**

OPEN

UP $\frac{(m-1)n(n-1)}{r-1}$: tail–head pair counting [p. 85]
LO $n(m-1)$ $n \geq r$: every tail gets $m-1$ copies, meeting the bound exactly at $n=r$ [p. 86]

$K_m^{\text{dir}(r)}(n)$ **vertex**

OPEN

UP $\frac{(m-1)n(n-1)}{r-1}$: carries over from the arc problem, and the general (any tail/head split) orientation shares the constant [p. 85,87,90]
LO $n(m-1)$ $n \geq r$: same construction [p. 86]

Erdős Problem 915: Cheat Sheet (complexity)

crucial running times for the max-flow checker and the exhaustive search built on it, citations into `program/`

One max-flow query

DFS Ford–Fulkerson, capped at $k+1$ paths $O(k n^2)$

dense residual scan (`for y in range(n)` per popped vertex, not an adjacency list), so one augmenting-path search alone costs $O(n^2)$ regardless of sparsity [`erdos915_unified.py:2300`, `_erdos_fast.c:20`]

local separation query, $k = m$: $\ell^{\max}(u, v)$ or $\kappa^{\max}(u, v)$ $O(m n^2)$

the vertex version splits every node in two ($2n$ nodes) for the internally-disjoint network, same exponent [`erdos915_unified.py:2352`]

Certifying one graph

$\ell^{\max}(G) \leq m - 1$ or $\kappa^{\max}(G) \leq m - 1$, all pairs $O(m n^4)$

$\Theta(n^2)$ pairs swept, each an $O(m n^2)$ query [`_erdos_fast.c:78` `max_connectivity_exceeds`, `erdos915_unified.py:2431` `feasible_after`]

The exhaustive search

branch-and-bound over every potential arc/edge $O(2^{\Theta(n^2)} m n^4)$ **worst case**

$n(n-1)$ ordered arcs (directed) or $\binom{n}{2}$ edges (undirected), each included/excluded, one $O(m n^4)$ feasibility recheck per node; the feasibility-monotonicity and bound prunes cut this hugely in practice but do not lower the worst-case exponent [`erdos915_unified.py:2374–2460`]

isomorphism reduction, base cases only $O(n! n^2)$

lex-min canonical form by Heap’s algorithm, $n \leq 7$ in practice [`_erdos_fast.c:97`]